

Deok-Hyeon Cho

Ph.D. Candidate

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🌐 <https://choddeok.github.io/Choddeok/>

🐙 <https://github.com/Choddeok>

🌐 <https://www.linkedin.com/in/deokhyeon-cho-7b522204>

🌐 <https://scholar.google.co.kr/citations?user=cynqcHwAAAAJ&hl=en>



Employment History

- 2022 – 2022 **RexSoft**, Developer Intern,
Participated in debugging and development of the REX program in the Development Department.
- 2021 – 2021 **Visang Education**, Data Scientist Intern,
Assisted with data entry, validation, analysis, and report creation for the E-Learning Content Planning Department.

Education

- 2022 – **Korea University**, Seoul, South Korea.
Integrated M.S.&Ph.D. in Artificial Intelligence in Pattern Recognition & Machine Learning Lab, under the supervision of Seong-Whan Lee. (GPA: 4.19 / 4.50)
- 2016 – 2022 **Hanyang University**, Ansan, South Korea.
B.S. degree in Applied Mathematics. (GPA: 4.40 / 4.50)

Research Publications and Patents

Research Publications

- 1 **D.-H. Cho**, S.-B. Kim, H.-S. Oh, and S.-W. Lee, “Affectflow: Word-level affective flow modeling for conversational speech synthesis,” in (*Under Review*), 2026.
- 2 **D.-H. Cho**, H.-S. Oh, S.-B. Kim, and S.-W. Lee, “Affectron: Emotional speech synthesis with affective and contextually aligned nonverbal vocalizations,” in *The 64th Annual Meeting of the Association for Computational Linguistics (ACL Findings)*, 2026.
- 3 H.-S. Oh, **D.-H. Cho**, S.-B. Kim, and S.-W. Lee, “L-proto: Language-aware episodic prototypical training for multilingual speaker verification,” in *Conference of the International Speech Communication Association (INTERSPEECH)*, 2026.
- 4 H.-S. Oh, **D.-H. Cho**, S.-B. Kim, and S.-W. Lee, “Toward complex-valued neural networks for waveform generation,” in *Proc. Conference of the International Conference on Learning Representations (ICLR)*, 2026.
- 5 **D.-H. Cho**, H.-S. Oh, S.-B. Kim, and S.-W. Lee, “Diemo-tts: Disentangled emotion representations via self-supervised distillation for cross-speaker emotion transfer in text-to-speech,” in *Proc. Conference of the International Speech Communication Association (INTERSPEECH)*, 2025.
- 6 **D.-H. Cho**, H.-S. Oh, S.-B. Kim, and S.-W. Lee, “Emosphere-ser: Enhancing speech emotion recognition through spherical representation with auxiliary classification,” in *Proc. Conference of the International Speech Communication Association (INTERSPEECH)*, 2025.
- 7 **D.-H. Cho**, H.-S. Oh, S.-B. Kim, and S.-W. Lee, “Emosphere++: Emotion-controllable zero-shot text-to-speech via emotion-adaptive spherical vector,” in *IEEE Transactions on Affective Computing (TAFAC)*, 2025.

- 8 N.-G. Kim, **D.-H. Cho**, S.-B. Kim, and S.-W. Lee, "Spotlight-tts: Spotlighting the style via voiced-aware style extraction and style direction adjustment for expressive text-to-speech," in *Proc. Conference of the International Speech Communication Association (INTERSPEECH)*, 2025.
- 9 H.-S. Oh, S.-H. Lee, **D.-H. Cho**, and S.-W. Lee, "Durflex-evc: Duration-flexible emotional voice conversion leveraging discrete representations without text alignment," in *IEEE Transactions on Affective Computing (TAFFC)*, 2025.
- 10 **D.-H. Cho**, H.-S. Oh, S.-B. Kim, S.-H. Lee, and S.-W. Lee, "Emosphere-tts: Emotional style and intensity modeling via spherical emotion vector for controllable emotional text-to-speech," in *Proc. Conference of the International Speech Communication Association (INTERSPEECH)*, 2024.
- 11 J.-E. Lee, S.-B. Kim, **D.-H. Cho**, S.-H. Lee, and S.-W. Lee, "Promoticon: Prompt-based emotion controllable text-to-speech via prompt generation and matching," in *Proc. IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2024.

Domestic Patents

- 1 "Apparatus and method for complex-valued neural network-based speech waveform generation," 10-2026-0106629, 2026.
- 2 "Method and system for emotional speech synthesis with affectively and contextually aligned nonverbal vocalizations," 10-2026-0067599, 2026.
- 3 "Apparatus and method for speech synthesis," 10-2025-0088028, 2025.
- 4 "Method and system for expressive text-to-speech via voiced-aware style extraction and style direction adjustment," 10-2025-0116457, 2025.
- 5 "Method for cross-speaker emotion transfer in text-to-speech using disentangled emotion representations via self-supervised distillation," 10-2025-0130127, 2025.
- 6 "Emotional expression voice generation apparatus and method capable of controlling emotional style and intensity using continuous emotional dimensions," 10-2024-0029066, 2024.
- 7 "Method, device, and program for synthesizing voices expressing emotions based on prompts," 10-2024-0099370, 2024.

Miscellaneous Experience

Awards, Achievements, and Scholarship

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| 2025 | <div style="display: flex; align-items: center;"> <div style="width: 10px; height: 10px; background-color: #800000; margin-right: 5px;"></div> <div> Excellence Award in Extreme-Noise Speech Recognition & Restoration AI Model Development Competition, Korea Artificial Intelligence Association (KAIA). </div> </div> |
| 2022 | <div style="display: flex; align-items: center;"> <div style="width: 10px; height: 10px; background-color: #800000; margin-right: 5px;"></div> <div> [BK21 Fellowship] Securing Outstanding Graduate Students, Korea University.
 <div style="display: flex; align-items: center;"> <div style="width: 10px; height: 10px; background-color: #800000; margin-right: 5px;"></div> <div> Department Prize for Outstanding Student Performance, Hanyang University. </div> </div> </div> </div> |
| 2021 | <div style="display: flex; align-items: center;"> <div style="width: 10px; height: 10px; background-color: #800000; margin-right: 5px;"></div> <div> Excellence Award in the Credit Card User Delinquency Prediction AI Competition, organized by Hanyang University on Dacon. </div> </div> |

Reviewer

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| 2024- | <div style="display: flex; align-items: center;"> <div style="width: 10px; height: 10px; background-color: #800000; margin-right: 5px;"></div> <div>IEEE International Conference on Systems, Man, and Cybernetics (SMC)</div> </div> |
| 2025- | <div style="display: flex; align-items: center;"> <div style="width: 10px; height: 10px; background-color: #800000; margin-right: 5px;"></div> <div>IEEE International Conference on Acoustics Speech and Signal Processing (ICASSP)</div> </div> <div style="display: flex; align-items: center;"> <div style="width: 10px; height: 10px; background-color: #800000; margin-right: 5px;"></div> <div>IEEE Transactions on Affective Computing (TAFFC)</div> </div> <div style="display: flex; align-items: center;"> <div style="width: 10px; height: 10px; background-color: #800000; margin-right: 5px;"></div> <div>IEEE Signal Processing Letters (SPL)</div> </div> |
| 2026- | <div style="display: flex; align-items: center;"> <div style="width: 10px; height: 10px; background-color: #800000; margin-right: 5px;"></div> <div>International Conference of the International Speech Communication Association (INTERSPEECH)</div> </div> |

Miscellaneous Experience (continued)

- 📖 Conference on Neural Information Processing System (NeurIPS)
- 📖 IEEE Spoken Language Technology Workshop (SLT)

Research Projects

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| 2024 - 2024 | Samsung Research , South Korea.
Development of interjection utterances for natural speech synthesis. |
| 2025 - 2025 | Thomas Crown , USA.
Voltron: Cross-speaker Controllable Emotion Transfer TTS. |
| 2025 - 2026 | Murf AI , USA.
Full Duplex Speech-to-Speech Conversational AI. |